

Review

Applications of thermochromic and electrochromic smart windows: Materials to buildings

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SUMMARY

Thermochromic and electrochromic smart windows are increasingly receiving attention for their specific ability to regulate the dynamics of light and heat. The goal of improving smart-window materials is chiefly to achieve building applications. This paper reviews the progress of existing material technologies and summarizes their experiments and simulations related to buildings. To facilitate the leap from material to building, the concept of performance regulation of ideal smart windows is proposed, and the applicability distribution of existing technologies is mapped. Based on the completely different core logic of thermochromic and electrochromic smart windows, the discrepancies and consistencies between the two are discussed. Performance requirements of smart windows in materials and buildings are proposed. Bridging the gap between materials and buildings for interdisciplinary promotion is the key to further exploration of the potential application of smart windows and accelerating their applications in the future.

INTRODUCTION

The building and construction industries account for almost one-third of total global energy consumption and approximately 15% of carbon dioxide emissions. Energy-efficient buildings are of substantial importance for achieving carbon neutrality, and green-certified office buildings can also bring a higher financial premium.^{1,2} Heating, ventilation, and air conditioning (HVAC) energy consumption comprises the largest proportion of building energy consumption,^{3–5} so it is important to reduce the building cooling and heating loads.⁶ Windows have worse heat transfer properties than walls, and about 60% of the heat loss in residential building structures can be attributed to glazed areas.⁷ As the proportion of building curtain walls⁸ increases, the energy-saving potential of windows cannot be ignored. Windows, as a transparent envelope component, have various functions, such as thermal insulation, lighting, and ventilation, and commonly have a remarkable impact on the light and thermal comfort of personnel, so optimizing window performance is necessary to enhance energy efficiency and guarantee the quality of the indoor thermal environment.

After the energy crisis in the 1970s, scholars conducted a lot of exploration of energy-saving glass. The most mature energy-saving glass in the market is low-emissivity (LowE) glass,^{9,10} but its adjustability is poor, and its application scenarios are limited. In the 1980s, the concept of “smart windows” was proposed by Svensson and Granqvist.¹¹ A smart window is an energy-saving window that adjusts solar radiation by combining dimming materials with glass and other substrates. Furthermore, various physical stimuli can trigger the variations in its optical properties to regulate

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the indoor environment, such as transmittance, reflectance, and absorption rate. Previous investigations on smart windows included thermochromic,^{12,13} electrochromic,^{14,15} gasochromic,^{16,17} mechanochromic,^{18,19} photochromic,^{20,21} and humidity chromic^{22,23} Windows. For various reasons, such as cost, performance, longevity, and scale, thermochromic and electrochromic technologies are anticipated to be implemented in the building field. Thermochromic windows (TCWs) and electrochromic windows (ECWs) have been commercialized, and ECWs are more mature. Representative companies related to the TCWs are Suntuitive from the United States and Magic Glass from the United Kingdom. Representative companies associated with ECWs are SageGlass, View Dynamic Glass, and ConverLight from the United States, Gesimat, and EControl-Glas from Germany.

Studies associated with smart windows were also added up by bibliometric methods, and the word cloud analyses in [Figure S1](#) and [Note S1](#) show that TCWs and ECWs have received more attention compared with other types of smart windows; however, very little attention has been paid to energy efficiency and building applications. Although the application feasibility of TCWs and ECWs has been preliminarily verified at the laboratory level, large-scale practical application still faces severe challenges.^{24,25} The materials field should address issues such as optimizing the dynamic regulation performance of windows, developing reliable large-scale fabrication technologies, reducing costs, and improving stability and lifetime. The building field should evaluate the suitability of smart windows in various climate zones, explore the best control strategy, and comprehensively evaluate the contribution of smart windows to energy efficiency, indoor environmental improvement, and personnel comfort. Implementation of smart windows requires cooperation of the materials and building fields, so it is necessary to review the research progress in both fields through a timely review and to bridge the gap to future applications. [Table 1](#) summarizes the reviews pertinent to smart windows over the past 5 years. These reviews have different emphases on the fields of materials and building, and most focus on materials. Some summarized emerging materials²⁶ and advanced integration technologies,²⁷ while some focused on a certain class of materials, such as mechanisms and properties of thermally responsive organic materials,¹² performance enhancement strategies²⁸ and fabrication conditions²⁹ for VO₂, and device design for hydrogels.³⁰ For applications of smart windows, the existing reviews are only mentioned or briefly discussed in the outlook.

Although these reviews may be an increasingly in-depth collation of research advances in the field of materials, the consideration of building applications is not deep and comprehensive enough. The final goal of technology research deployment in practical applications, so the development of materials technology for smart windows must be appropriately matched with building applications. Only a few individual reviews have discussed application performance. Cui et al.,³³ for instance, referred to VO₂-based building simulations and small-chamber experiments and emphasized the energy efficiency of VO₂-based smart windows but did not evaluate the built environment and human comfort that must be met for practical application. Aburas et al.⁸ summarized the progress of building simulation work on TCWs and noted that existing simulation studies have focused on the performance of various building applications, reflecting the need for a more comprehensive assessment of the effects of various types of smart windows from an application perspective. Overall, the existing reviews are still insufficient to consider practical application of smart windows. With the rapid development of related technologies, they should be reviewed in time to realize a smooth transition from material to building.

Table 1. Review articles on smart windows in the last 5 years

Reference	Type of smart window	Material	Building
Cai et al. ³¹	EC	mainly introduced (preparation methods)	brief discussion (large-scale manufacturing, cost)
Li et al. ²⁸	thermochromic (VO ₂)	mainly introduced (performance improvement strategies)	brief discussion (large-scale preparation requirements)
Granqvist et al. ³²	EC	mainly introduced (principles and materials)	less introduced (long-term durability assessment)
Ke et al. ²⁷	thermochromic	mainly introduced (emerging materials)	brief discussion (development prospect)
Cui et al. ³³	thermochromic (VO ₂)	mainly introduced (performance improvement strategies)	less introduced (practical energy efficiency estimated)
Ke et al. ²⁶	EC, thermochromic, mechanochromic, photochromic	mainly introduced (functional materials and equipment design)	mentioned only
Aburas et al. ⁸	thermochromic	less introduced (optical film data)	mainly introduced (energy-saving performance and building simulation studies)
Vu et al. ²⁹	thermochromic (VO ₂)	mainly introduced (physical vapor deposition techniques)	brief discussion (large-scale manufacturing)
Zhou et al. ³⁰	thermochromic, EC, humidity-chromic (hydrogel)	mainly introduced (mechanisms and fabrication methods)	mentioned only
Zhou et al. ⁶	humidity-chromic, mechanochromic, magneto-chromic	mainly introduced (structures and designs)	mentioned only
Feng et al. ¹²	thermochromic	mainly introduced (mechanisms and performance improvement strategies)	brief discussion (large-scale manufacturing, cost)

This paper aims to provide new ideas for evaluating the performance of smart windows from the perspective of practical application and promote integration of smart windows from materials to buildings. This review introduces the development of material technologies for TCWs and ECWs and the experimental and simulation research related to building applications. Additionally, the concept of ideal performance of smart windows for buildings is presented, and the climate suitability distribution of existing technologies is mapped appropriately. The discrepancies and consistencies between TCWs and ECWs are discussed in detail based on their different core logics. The performance requirements for development of smart windows in materials and buildings are further discussed. It is hoped that this review will help the materials field to clarify the future research goals of smart windows, explore their application potential, and promote the interdisciplinary intersection of materials and buildings.

TCWs

TCWs in material development

Thermochromic processes can be achieved by a variety of thermo-responsive materials, including VO₂, hydrogels, ionic liquids, liquid crystals, and perovskites. Perovskites and liquid crystals essentially modulate the transmission of visible and

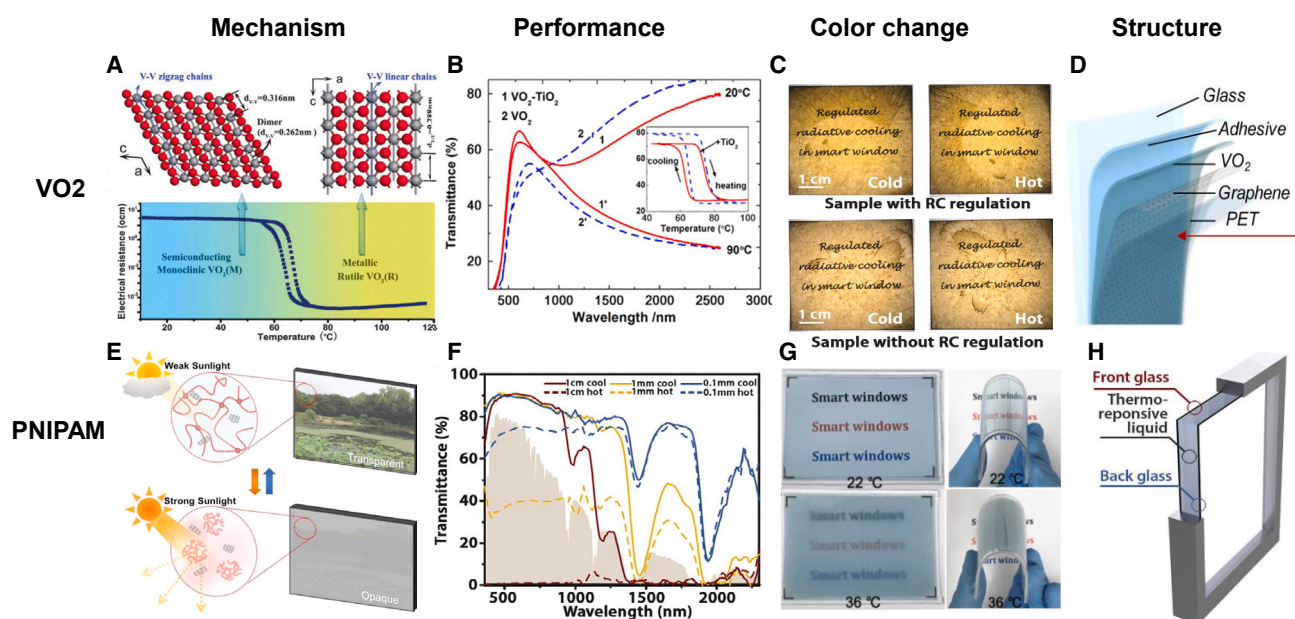


Figure 1. Thermochromic mechanism, performance, color, and smart window structure of VO₂ and PNIPAM

(A) Mechanism of VO₂ phase transition.³⁸
 (B) Transmission spectra of VO₂ samples at 20°C and 90°C.³⁵
 (C) Passive radiative cooling (RC) regulating TCW colors at different temperatures.³⁹
 (D) Structure and sample photo of a smart window coated with a VO₂ layer.⁴⁰
 (E) Mechanism of hydrogel phase transition.⁴¹
 (F) Transmission spectra of thermo-responsive liquid with various thicknesses at 20°C (solid line) and 60°C (dashed line).⁴²
 (G) Transparency of Cs_xWO₃/PAM-PNIPAM flexible films before and after phase transition temperature.⁴³
 (H) Structure and sample photos of the high-energy storage thermo-responsive smart window.⁴²
 Reproduced with permission from (A) Wu et al.,³⁸ Royal Society of Chemistry; (B) Chen et al.,³⁵ Elsevier; (C) Wang et al.,³⁹ The American Association for the Advancement of Science; (D) Kim et al.,⁴⁰ American Chemical Society; (E) Kim et al.,⁴¹ Springer Nature; (F) and (H) Zhou et al.,⁴² Elsevier; (G) Wu et al.,⁴³ American Chemical Society.

near-infrared (NIR) light, respectively; their solar modulation range is limited, and, in particular, their phase change temperature (t_c) is not appropriate for building applications.³⁴ Studies of on VO₂ and hydrogels are more typical and mature, so this paper primarily reviews their corresponding related technologies.

Figure 1A demonstrates the reversible phase transition between the metal and semiconductor of VO₂ at 68°C. At room temperature, VO₂ represents a monoclinic (M) system in the semiconductor state with high infrared transmittance, and above the phase transition temperature, its crystal structure is altered to a rutile (R) tetragonal system in the metal state with high infrared reflectance, while luminous transmittance (T_{lum}) remains relatively unchanged.^{35,36} Figure 1B also shows that the modulation range of VO₂ on the solar spectrum is mostly placed in the NIR band.³⁵ As shown in Figure 1C, the intrinsic color of VO₂ is yellowish, but the color of the film sample is also related to the solution preparation method, coating concentration, and structural layer of the sample. Commonly, VO₂ is prepared by physical vapor deposition methods or solution processing approaches³⁷ and overcoated with other functional layers on a transparent substrate to form the TCW samples (Figure 1D).

Compared with VO₂, poly(N-isopropylacrylamide) (PNIPAM) hydrogel exhibits a lower phase transition temperature. It experiences phase transition near 33°C and transforms from a homogeneous system to a non-homogeneous one. As can be seen in Figure 1E, below the lower critical solution temperature (LCST), PNIPAM

aqueous solution possesses high transparency, and most of the solar radiation can be transmitted. When it is heated above the LCST, the water molecules are released from PNIPAM, and the shrinkage particles will cause scattering of light.⁴⁴ Figure 1F illustrates that PNIPAM hydrogels modulate the solar spectrum in the visible and NIR portions. As the temperature increases, the transparency of the thermally responsive liquid gradually lessens. When other components are added to the configuration solution, the hydrogel may have a certain color; Figure 1G shows a thermochromic flexible film of Cs_xWO₃/polyacrylamide (PAM)-PNIPAM⁴³ from light blue transparent to atomized. The transparency depends on microgel size and hydrogel thickness.⁴⁴ Hydrogel is commonly sandwiched between two layers of clear glass to configure smart windows, as shown in Figure 1H.

T_{lum} and solar modulation ability (ΔT_{sol}) are two crucial indicators⁴⁵ to characterize the performance of thermochromic materials, where T_{lum} denotes the transmittance in the wavelength range of 380–780 nm; window materials should have the highest possible visible light transmittance to ensure good interior illumination. ΔT_{sol} represents the discrepancy between the transmittance of the entire solar spectrum at low and high temperatures and reflects the energy efficiency of the material. These two factors are defined in the following form:

$$T_{lum} = \frac{\int_{380}^{780} \varphi_{lum}(\lambda) T(\lambda) d\lambda}{\int_{380}^{780} \varphi_{lum}(\lambda) d\lambda} \quad (\text{Equation 1})$$

$$T_{sol} = \frac{\int_{250}^{2500} \varphi_{sol}(\lambda) T(\lambda) d\lambda}{\int_{250}^{2500} \varphi_{sol}(\lambda) d\lambda} \quad (\text{Equation 2})$$

$$\Delta T_{sol} = T_{sol,b} - T_{sol,c} \quad (\text{Equation 3})$$

where $\varphi_{lum}(\lambda)$ is the standard luminous efficiency function, $\varphi_{sol}(\lambda)$ is the solar irradiance spectrum of air mass 1.5, $T(\lambda)$ is the spectral transmittance, and $T_{sol,b}$ and $T_{sol,c}$ are the solar radiation transmission rates of TCWs at low temperatures (transmittance state) and high temperatures (tinted state), respectively. The value of ΔT_{sol} for monolayer VO₂ films is typically below 10%,⁴⁶ and ΔT_{sol} of existing VO₂ technologies rarely reaches 30%. To achieve building applications, the primary conditions of ΔT_{sol} of 50%³³ and T_{lum} of 60%^{37,47} should be satisfied, but this range is only a compromise expectation. In practical applications, the values of T_{lum} and ΔT_{sol} depend on the climate zones and building types in which the smart windows are used. The material performance affects the application performance, including energy efficiency, indoor light and thermal environment, and personnel perception and comfort. Building simulations can be employed to analyze the range of material properties that can meet the building applications with the actual application requirements as the target. Thermochromic technologies associated with VO₂ focus on overcoming three difficulties: (1) lowering the t_c to near room temperature, (2) increasing the T_{lum} , and (3) improving the modulation of the solar spectrum.⁴⁸ The main techniques for enhancing the properties of VO₂ and their characteristics are provided in Table S1, including elemental doping,^{39,49} nanocomposites,^{50,51} porous films,⁵² grid structures,⁵³ biomimetic VO₂ patterning,⁵⁴ antireflection coating,^{35,55,56} and multi-layered structures.^{57,58} Among them, nanocomposites are probably the best approach to balance T_{lum} and ΔT_{sol} .

PNIPAM hydrogels are essentially prepared by chemical solutions, and hydrogel thermochromic materials exhibit better visible light transmission and solar spectrum modulation than VO₂ films (Figure S2). The challenges faced by PNIPAM hydrogels

include (1) evaporation and freezing of water molecules, (2) deformation because of shrinkage and leakage of the phase change process, and (3) structural stability of the microgel. Related performance enhancement strategies are shown in [Table S2](#), such as enhancing modulation capabilities,^{59–61} strengthening structural stability,^{42,59,62,63} and stimulating phase changes.^{41,43,64} The combined effect of the materials enhances the modulation ability of the hydrogel,⁵⁹ while SiO₂ can be exploited as a protective layer,⁶³ or the microgel can be locked in a polymer lattice to enhance mechanical stability and durability.⁶² Polymers that are sensitive to sunlight and temperature are prepared by combining materials such as graphene with hydrogels to induce hydrogel phase transitions.^{43,60,64}

TCWs in building applications

Chamber experiments

Because of the complex production process and high manufacturing cost of smart window products, they are not yet available for large-scale application. To validate the feasibility of smart window technologies, researchers typically build several small model chambers, using ordinary glass and commercial LowE glass as samples to compare their environment-built performance with smart window samples. Air temperature in the chamber and window surface temperature are commonly utilized as measurement indicators. Some representative small-chamber experiments are summarized in [Table 2](#) and [Figure 2](#).

As shown in [Table 2](#), small model chambers with various window samples installed are typically placed in a controlled static environment without temperature fluctuations. An infrared light source is exploited as a solar simulator and placed outside of the side window ([Figure 2A](#))⁴² or skylight ([Figure 2B](#)).⁴³ The surface and interior temperatures rise as windows are heated, but the TCWs suppress the temperature rise and exhibit better thermal insulation because their transmittance changes as the surface temperature rises to trigger the phase transition. Typical results are illustrated in [Figure 2A](#); the interior temperatures were 57°C and 34°C for ordinary glass and hydrogel smart windows (with a temperature difference of 23°C), respectively, reflecting the huge energy-saving potential of the smart windows. In addition to the infrared light source, Lin et al.⁶² simulated daytime experiments in winter and summer by employing irradiated lamp sources, and conducted nighttime experiments with heaters and coolers, showing that their proposed smart window makes the room warmer in winter and cooler in summer compared with other windows, fully verifying the applicability of the proposed smart window at various time intervals ([Figure 2C](#)). [Figure 2D](#) illustrates a large-scale outdoor experiment with VO₂ glass installed,⁶⁵ indicating that the feasibility study of smart windows is gradually moving from a controlled environment to an actual outdoor one, which provides important support for more extensive application in the future. It should be emphasized that the discrepancies between the application performance of indoor models and real buildings are still worth further discussion, and more attention should be paid to the effect of the indoor environment on the visual field and the comfort of personnel.

Building simulations

Researchers have prepared thermochromic films and window samples in the laboratory and tested their spectral properties. In addition to building small-chamber models to verify the effect of TCWs on indoor environments, the spectral properties of the samples are imported into software for further building simulations, where building dimensions and window-wall ratios (WWRs) are modified to explore their applicability in a variety of scenarios. There is no doubt that validating the functional performance of TCWs through building simulation is the reason and motivation for

Table 2. Some existing experiments relevant to thermochromic chambers

Reference	Material	t_c (°C)	Experimental environment	Dimensions of chamber	Window orientation	Window size	Time
Gao et al. ⁵⁵	VO ₂ -Sb:SnO ₂	N/A	Static room at 15°C, heating with 150-W infrared lamp	34 × 27 × 29 cm	side window	25 × 30 cm	700 s
Ye et al. ⁶⁵	VO ₂	41.3	outdoor	2.9 × 1.8 × 1.8 m	side window	1.65 × 1.65 m	N/A
Zheng et al. ⁵⁷	TiO ₂ /VO ₂ /TiO ₂	61.5	static room at 13°C, heating with 150-W IR lamp	55 × 36 × 43 cm	side window	40 × 40 cm	90min
Wu et al. ⁴³	Cs _x WO ₃ /PNIPAM	30	static room at 22°C with 100 W/m ² irradiation	N/A	skylight	N/A	60 min
Li et al. ⁶⁰	PNIPAM/2-aminoethylmethacrylate hydrochloride (AEMA)	31	static room at 25°C with 100 mW/cm ² irradiation	15 × 10 × 8 cm	side window	4 × 5 cm	30 min
Zhou et al. ⁴²	PNIPAM	32.5	static room at 24°C, heating with 500-W infrared lamp	20 × 20 × 30 cm	side window	20 × 20 cm	3,000 s
			outdoor in January in Singapore	20 × 20 × 30 cm	skylight	20 × 20 cm	12:00
			outdoor in December in Beijing	20 × 20 × 30 cm	skylight	20 × 20 cm	24:00
Wang et al. ⁶⁶	HPC hydrogel	31	outdoor in January in Singapore	20 × 20 × 30 cm	skylight	10 × 10 cm	4 h
			static room at -7°C, heating with 100-W IR lamp	20 × 20 × 30 cm	skylight	10 × 10 cm	90 min
Zhang et al. ⁶⁷	HPC/PAA hydrogel	26.5	static room at 22°C, heating with 700-W IR lamp	30.5 × 26 × 27 cm	side window	N/A	60 min
Lin et al. ⁶²	PNIPAM/AgNW	31	static room at 5°C with 800 W/m ² irradiation (winter day)	12 × 12 × 15 cm	side window	N/A	40 min
			static room at 28°C with 1,000 W/m ² irradiation (summer day)	12 × 12 × 15 cm	side window	N/A	40 min
			static room at 6°C with a 3-W heater (winter night)	12 × 12 × 15 cm	side window	N/A	120 min
			static room at 30°C with a 13-W cooler (summer night)	12 × 12 × 15 cm	side window	N/A	60 min

N/A, not available.

the success of materials development. Building simulation can also assume window performance parameters and perform sensitivity analyses to uncover the best way to enhance the performance of smart windows. Thus, it can be implemented as a reverse performance-oriented approach for further efforts in building applications.

There are various software tools for TCW modeling.⁴⁷ In a literature review by Aburas et al.,⁸ simulation tools for TCWs are introduced and compared, noting that EnergyPlus is the most commonly used software for exploring the effects of TCWs on building applications. EnergyPlus integrates the thermochromic module "WindowMaterialGlazingGroup: Thermochromic," which determines the temperature of the glass at each time step and assigns the most suitable optical data to the glass, finely characterizing the thermochromic properties of the material. EnergyPlus can export year-round energy consumption data, hourly indoor temperature, and light profiles for a comprehensive assessment of energy savings, visual comfort, and thermal comfort.

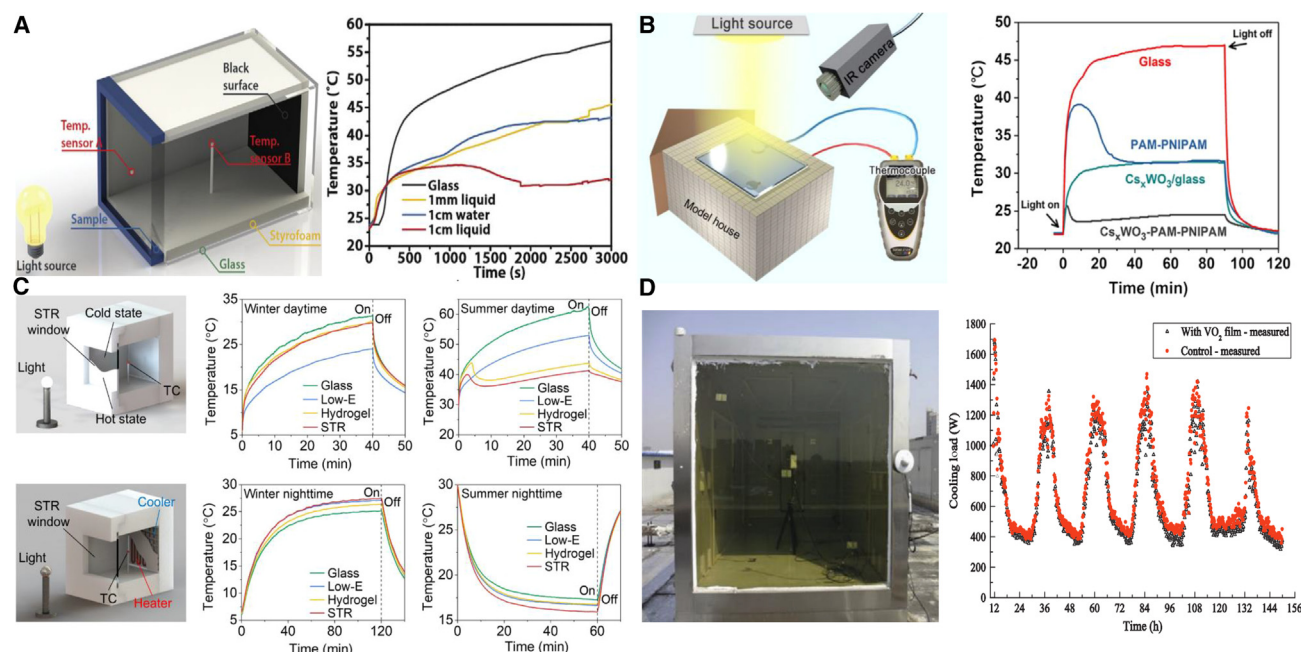


Figure 2. Small-chamber experiments related to thermochromic technologies

(A) Comparison of room temperatures for various thicknesses of PNIPAM thermally responsive liquid and normal glass as side windows.⁴²

(B) Comparison of room temperature between PNIPAM hydrogel and normal glass as skylights.⁴³

(C) Performance comparison of smart windows with ordinary glass and LowE glass in winter, summer, day, and night.⁶²

(D) Outdoor experiments with VO₂ single-pane glass.⁶⁵

Reproduced with permission from (A) Zhou et al.,⁴² Elsevier; (B) Wu et al.,⁴³ American Chemical Society; (C) Lin et al.,⁶² The American Association for the Advancement of Science; (D) Ye et al.,⁶⁵ Elsevier.

It should be emphasized that the energy efficiency of TCWs depends not only on their material properties but also largely on the climatic conditions in which they are applied. Exploring the building suitability of TCWs should first start with climatic conditions to put demands on material selection for smart windows. Table 3 shows simulation studies of TCWs through EnergyPlus, showing that existing studies have explored the actual energy-saving performance of TCWs in different climate zones and orientations, finding that TCWs are more applicable in warmer climates because of their higher t_c . In addition to energy consumption metrics, studies have also focused on visual comfort,⁶⁸ lighting performance,^{69,70} and thermal comfort.^{71,72} To implement TCWs for building applications, future building simulation investigations should take into account energy efficiency performance, built environment, and occupant comfort in an integrated manner.

ECWs

ECWs in material development

Electrochromic materials are materials that alter color reversibly and autonomously through redox reactions because of externally alternating voltages, including inorganic transition metal oxides represented by WO₃⁷⁹ and organic conductive polymers.⁸⁰ The main advantage of these materials is their adjustability in transmitting sunlight according to the application demands.³² Electrochromic nanomaterials prepared by solution approaches could also open up new opportunities in terms of manufacturability and material properties.^{81,82}

The generic structure of electrochromic devices is represented by five superimposed layers on a single transparent substrate or located between two transparent

Table 3. Several building simulation studies of TCWs through EnergyPlus

Reference	Material	Location	Building type	Building size	WWR	Window orientation	$T_{lum,b}^a$; $T_{lum,c}^b$	$T_{sol,b}^c$; $T_{sol,c}^d$	K_b^e ; K_c^f	Concerns	Specific indicators
Hoffmann et al. ⁷³	hypothetical theoretical spectrum	Chicago; Houston	12-story office building	73 × 49 × 4 m for each layer	0%–60%	four cardinal directions	0.53; 0.27	0.39; 0.08	1.45; 1.45	energy consumption; visual comfort	accumulated annual window heat gain and heat loss; annual site energy use intensity; maximum number of possible annual hours
Warwick et al. ⁷⁴	hypothetical theoretical spectrum	Helsinki; Palermo; London	simple office room	6 × 5 × 3 m	100%	southern face	N/A	N/A	N/A	energy consumption	the total energy requirement and the breakdown of energy usage
Favoino et al. ⁷⁵	hypothetical theoretical spectrum	Helsinki; London; Rome	A typical office room	3 × 5 × 3.5 m	40%	four cardinal directions	N/A	N/A	N/A	energy consumption	specific total primary energy use; energy-saving potential
Allen et al. ⁷⁶	HPC hydrogel-based TCWs	Palermo	small office test cell	5 × 4 × 3 m	25%	southern face	0.90; 0.16	0.74; 0.11	2.7; 2.7	energy consumption	window heat gain; energy-saving potential
Liang et al. ⁷⁷	iron-liquid and VO ₂ -based film	Beijing; Guangzhou; Shanghai	small cellular office	6 × 5 × 3 m	60%	southern face	0.656; 0.605	0.692; 0.571	2.7; 2.7	energy and day lighting performance	the breakdown of energy usage; UDI _{500–2,000}
Zhou et al. ⁴²	PNIPAM hydrogel-based TCWs	Shanghai; Las Vegas; Riyadh; Singapore	simple office room	8 × 6 × 2.7 m	28% and 37%	four cardinal directions	0.923; 0.023	0.716; 0.016	3.23; 3.23	energy consumption	energy-saving potential; monthly HVAC energy and annual total energy
Zhou et al. ⁷⁸	VO ₂ -based TCWs	Athens; Guangzhou; Miami	N/A	N/A	N/A	N/A	0.433; 0.344	N/A	N/A	energy consumption	energy-saving potential; monthly HVAC energy and annual total energy
Zhao et al. ⁷¹	PMMA-VO ₂ based film	Boulder; Phoenix; Boston	N/A	N/A	N/A	southern face	0.50	N/A	N/A	energy consumption; thermal comfort	heating and cooling loads; inside surface temperature of the window
Aburas et al. ⁵⁶	VO ₂ -based TCWs	Cairo; Barcelona; London	small cellular office	5 × 4 × 3 m	60%	southern face	0.752; 0.721	0.768; 0.748	N/A	energy and day lighting performance	energy-saving; annual energy consumption; UDI _{500–2,000}
Wang et al. ⁶⁶	HPC hydrogel-based TCWs	Cairo; Shanghai; Madrid; Denver	12-story office building	73 × 49 × 4 m for each layer	40%	four cardinal directions	0.716; 0.078	0.577; 0.073	2.25; 1.98	energy consumption	monthly HVAC energy and annual total energy
Lin et al. ⁶²	PNIPAM hydrogel-based TCWs	Anchorage; Beijing; Hong Kong; Abu Dhabi	simple office room	8 × 8 × 3 m	12.5% and 33%	four cardinal directions	0.783; 0.100	0.655; 0.071	2.88; 4.00	energy consumption	monthly HVAC energy and annual total energy
Liu and Wu ⁶⁹	HPC hydrogel-based BIPV window	Nottingham	small cellular office	3 × 3 × 3 m	25%, 50%, and 75%	vertical window/skylight	0.698; 0.074	0.520; 0.055	N/A	energy and day lighting performance	monthly window solar heat gain; the breakdown of energy usage; UDI _{500–2,000}

^aLight transmittance of ECWs under a bleached state (thermochromic materials at low temperatures).

^bLight transmittance of ECWs under a colored state (thermochromic materials at high temperatures).

^cSolar transmittance of ECWs under a bleached state (thermochromic materials at low temperatures).

^dSolar transmittance of ECWs under a colored state (thermochromic materials at high temperatures).

^eHeat transfer coefficient under a bleached state.

^fHeat transfer coefficient under a colored state.

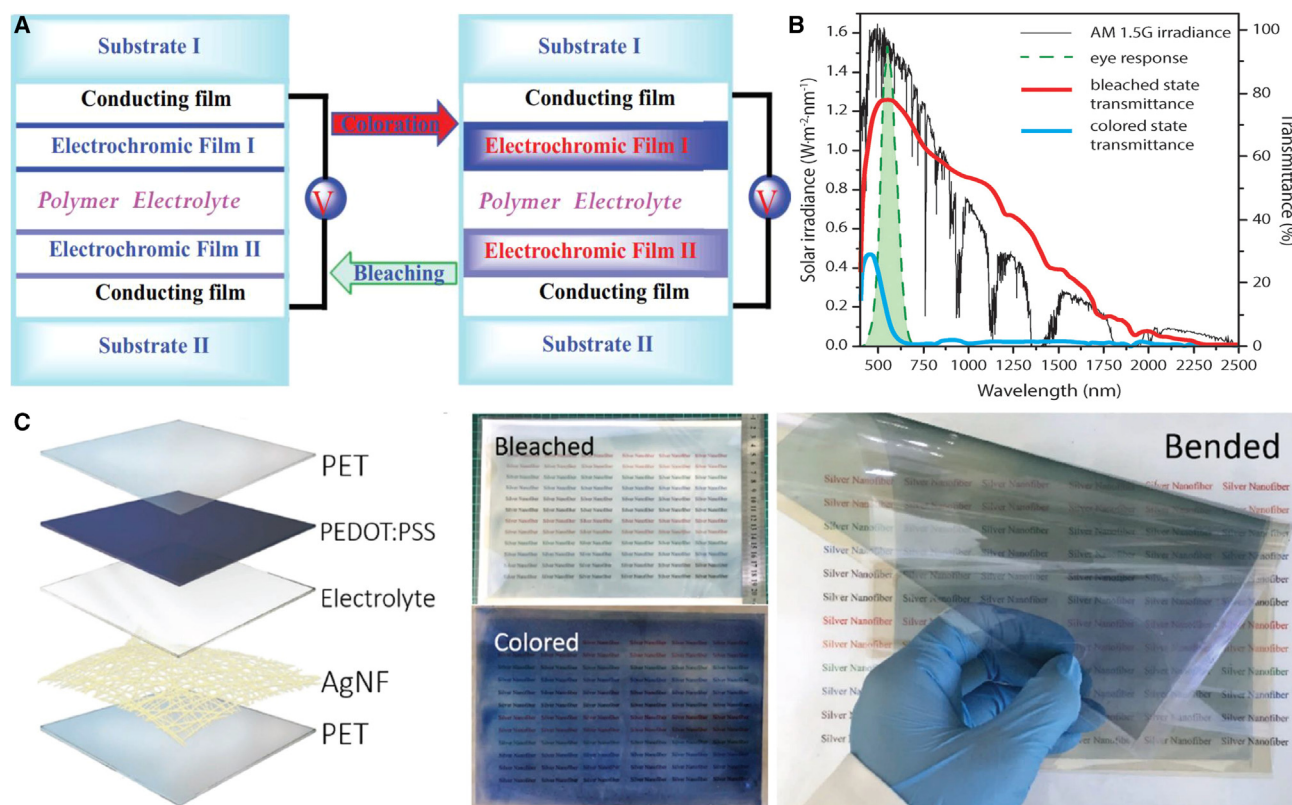


Figure 3. EC mechanism

(A) General construction of EC devices.⁸⁵

(B) Transmission spectra of EC devices subjected to bleaching and coloring.⁸¹

(C) Flexible EC devices.⁸⁰

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substrates. The five-layered structure consists of a transparent conductive layer coated with an ion storage film, a transparent conductive layer coated with an electrochromic layer, and an electrolyte layer sandwiched in the middle.⁸³ The electrochromic layer exhibits bleaching and coloring properties through redox reactions, while the ion storage layer serves to store and provide the ions required for the electrochromic material. The ion storage membrane can also be an electrochromic layer, as demonstrated in Figure 3A. The electrolyte layer acts as an ion transport channel and guarantees fast color switching. All constituent components must be highly transparent to visible light and electrochemically stable over the switching voltage range.⁸⁴ The transmission spectra of electrochromic devices in the presence of bleaching and coloring are presented in Figure 3B. Additionally, the flexible electrochromic devices based on the silver nanofiber (AgNF) transparent electrodes shown in Figure 3C can be fabricated on a large scale.⁸⁰

The visible light transmittance, coloration efficiency (CE), switching time, and demonstrated cycles of electrochromic materials in bleached and colored states represent crucial performance indicators in the field of ECWs.^{80,83} The CE⁸⁶ is defined as the ratio of the change of optical density at a specific wavelength to the insertion or extraction of charge per unit area of the film, which is measured in $\text{cm}^2 \text{C}^{-1}$. Therefore,

$$CE = \frac{\Delta OD}{Q} \quad (\text{Equation 4})$$

$$\Delta OD = \log(T_b / T_c) \quad (\text{Equation 5})$$

where ΔOD represents the change in optical density; T_b and T_c are the transmittance factors in the presence of bleached and colored states at a particular wavelength, respectively; and Q is the electron charge per unit area injected into the electrochromic material. Materials with higher CE will have better durability and faster switching times. The switching time is generally defined as the time required to complete 90% of the optical change between the stable bleached and colored states. To achieve implementation of electrochromic technology in buildings, technology development in this field is focused on the following four items: (1) the highest visible light transmittance possible; (2) improvement in response time; (3) durability, flexibility, and life span; and (4) integration with power generation, energy storage systems, and other energy-saving technologies. The technical developments are presented in Table S3. Electrochromic bleached states have higher transmittance, while tinted states are mostly blue and have reduced transmittance. The color of the window will affect visual comfort. Huang et al.⁸⁷ demonstrated ECWs for realizing multiple colors. Most existing technologies are capable of converting colors in a matter of seconds, which is sufficient for building applications. Durability commonly refers to the ability of electrochromic devices to undergo tens of thousands of cycles without significant loss of performance,⁸⁸ and the cycling performance should face the test of decades of building application. Electrochromic technology can be combined with membrane-like or even curvy structures to achieve more energy-saving potential, while flexible large-area preparation technology provides new opportunities in this field.^{89,90}

ECWs in building applications

Case studies

Electrochromic-related technologies have extensive potential for development in various fields. As demonstrated in Figure 4, many studies have been devoted to the research and development of optical thin film coating in the 1970s. In the 1980s, small-area sample experiments gradually emerged, and the advantages of its application in buildings and automobile rear-view mirrors were researched and discovered in the 1990s. With the progress of electronic information technology, it has shown more applications in flexible electronics and smart glasses. The organic electrochromic technology makes it easier to produce curved products, but its durability and longevity limit the possibility of long-term application. The most applicable technology is the inorganic electrochromic technology represented by WO_3 . Several international companies have produced ECWs to address application challenges such as large-area preparation, coloring time acceleration, stability, and longevity. ECWs have also been applied in practical buildings in the United States, France, and the Netherlands⁹¹ but are still a long way from extensive utilization. Field tests and examinations of the use of full-size spaces with electrochromic (EC) glass are still rare, and there are some scale model explorations, which provide a valuable reference experience for the application of ECWs, as compiled in Table 4.

Case studies with ECWs focused on the following aspects: energy demand reduction and improvement of the indoor environment,^{100,101,103} glare control,¹⁰⁴ and behavioral modulation preferences and perceptions of personnel when ECWs are combined with louvers and adjustable luminaires.^{98,99,102} Lee et al.⁹⁷ provided the first comprehensive demonstration of large-area ECWs in United States offices, noting that assessing subjective human responses and perceptions of dynamic window brightness and color, as

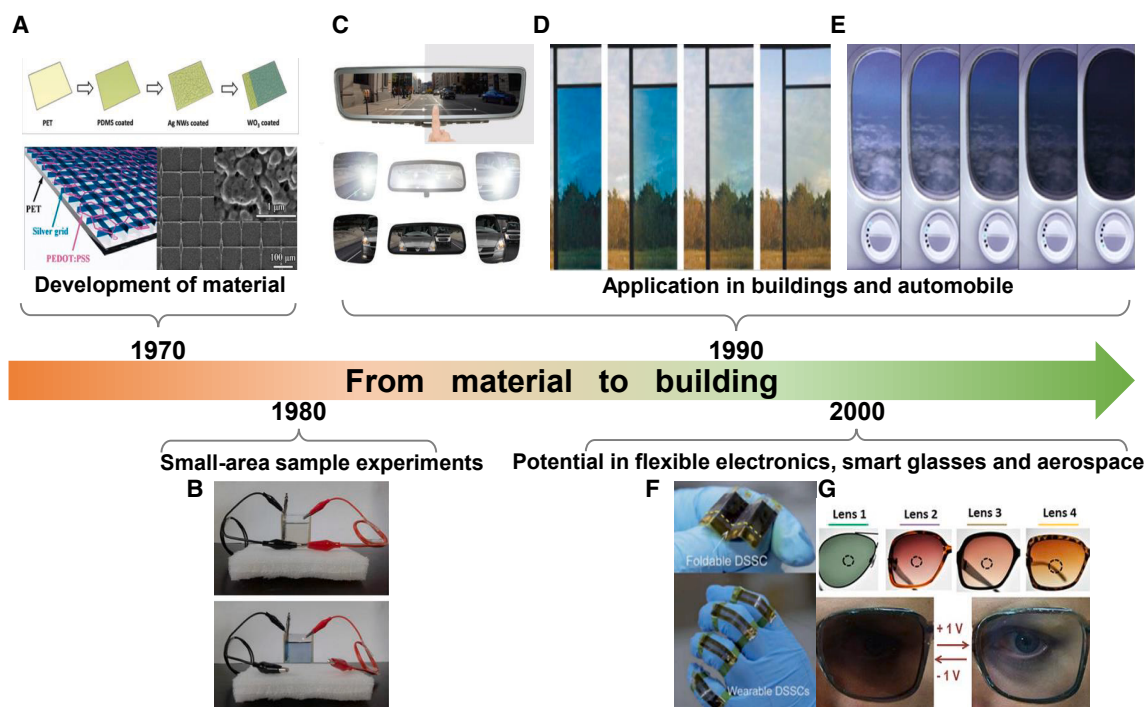


Figure 4. EC technology field

(A) Preparation of EC devices.⁹²

(B) Small-area sample experiments.⁹³

(C) Auto-dimming and anti-glare rearview mirror by Gentex.

(D) EC architectural glass from Gesimat.²⁴

(E) EC glass from SmartTint for Boeing aircraft.⁹⁴

(F) Flexible wearable electronics.⁹⁵

(G) EC glasses.⁹⁶

Reproduced with permission from (A) Lee et al.,⁹² John Wiley and Sons; (B) Wang et al.,⁹³ Springer Nature; (D) Baetens et al.,²⁴ Elsevier (E) Yang et al.,⁹⁴ Elsevier; (F) Zhou et al.,⁹⁵ Royal Society of Chemistry; (G) Österholm et al.,⁹⁶ American Chemical Society.

well as thermal comfort, are priorities for future research. In a later experiment, Clear et al.,⁹⁸ from the same team, further evaluated the behavioral regulatory preferences of 43 subjects in an office equipped with switchable ECWs, manual blinds, and dimmable fluorescent lamps (Figure 5A), showing that ECWs reduced the incidence of glare and were preferred over blinds. Jain et al.¹⁰⁴ assessed glare perception with blue ECWs in 20 participants exposed to different solar heights, confirming a strong angular dependence of glare perception, as illustrated in Figure 5B.

Solar protection in the tinted mode of EC glass can reduce cooling energy consumption, but the low transmittance in the tinted mode may enhance lighting energy consumption. To balance indoor heat gain and lighting energy consumption, SageGlass has proposed an interesting methodology of utilizing various tints of glass within the pane group.¹⁰⁵ As presented in Figure 5C, zone 1 is set to transparent ($T_{lum} = 60\%$), while zones 2 and 3 are set to tint ($T_{lum} = 1\%$). The day lighting in the space will be dominated by transparent glass, so this approach provides sufficient illumination, ensures that the lighting spectrum remains neutral in the space, and allows maximum energy savings. In practical application, the partition control of glass can not only avoid the necessity of large-area glass preparation but also improve the color change rate and optimize the color uniformity of each glass, so it aspires to achieve dual benefits of energy savings and comfort.

Table 4. Full-size room and scale model experiments associated with ECWs

Reference	Glass company	Test site	Room size	Window orientation	Window size	$T_{lum,bi}$ $T_{lum,c}$	$T_{sol,bi}$ $T_{sol,c}$	K_{bi} K_c	Switching time	Number of subjects
Lee et al. ⁹⁷	N/A	USA	45 m ³	southeast	6.69 m ²	0.38; 0.11	N/A	N/A	10–25 min	no
Clear et al. ⁹⁸	N/A	USA	47 m ³	south	6 m ²	0.60; 0.03	N/A	N/A	several minutes	43
Zinzi ⁹⁹	Flabeg	Italy	40 m ³	west; north	1.9 m ²	0.50; 0.15	0.36; 0.12	1.1; 1.1	12 min	30
Piccolo ¹⁰⁰	K-glass	Italy	0.08 m ³	south; west	0.014 m ²	0.46; 0.03	0.53; 0.12	2.2; 2.2	N/A	no
Ciampi et al. ¹⁰¹	Gesimat	Italy	198 m ³	south	2.4 m ²	0.725; 0.607	0.651; 0.554	2.5; 2.5	N/A	no
Bakker et al. ¹⁰²	N/A	the Netherlands	79 m ³	west	N/A	N/A	N/A	N/A	N/A	26
Frattolillo et al. ¹⁰³	EControl	Italy	55 m ³	south	1.3 m ² ; 0.57 m ²	0.62; 0.02	0.48; 0.10	1.1; 1.1	10 min	no
Jain et al. ¹⁰⁴	N/A	Switzerland	53 m ³	south	5 m ²	0.56; 0.006	N/A	N/A	N/A	20
Ajaji and André ⁶⁸	N/A	Belgium	0.79 m ³	south	0.5 m ²	0.463; 0.15	0.315; 0.15	1.25; 1.27	N/A	no

Building simulations

For TCWs, as fully passive windows, the external climate directly affects the window exterior surface temperature, which further determines the solar radiation transmittance. In contrast, for ECWs, as active modulating elements, the color change logic can be set artificially, and thus their control strategies determine their application.

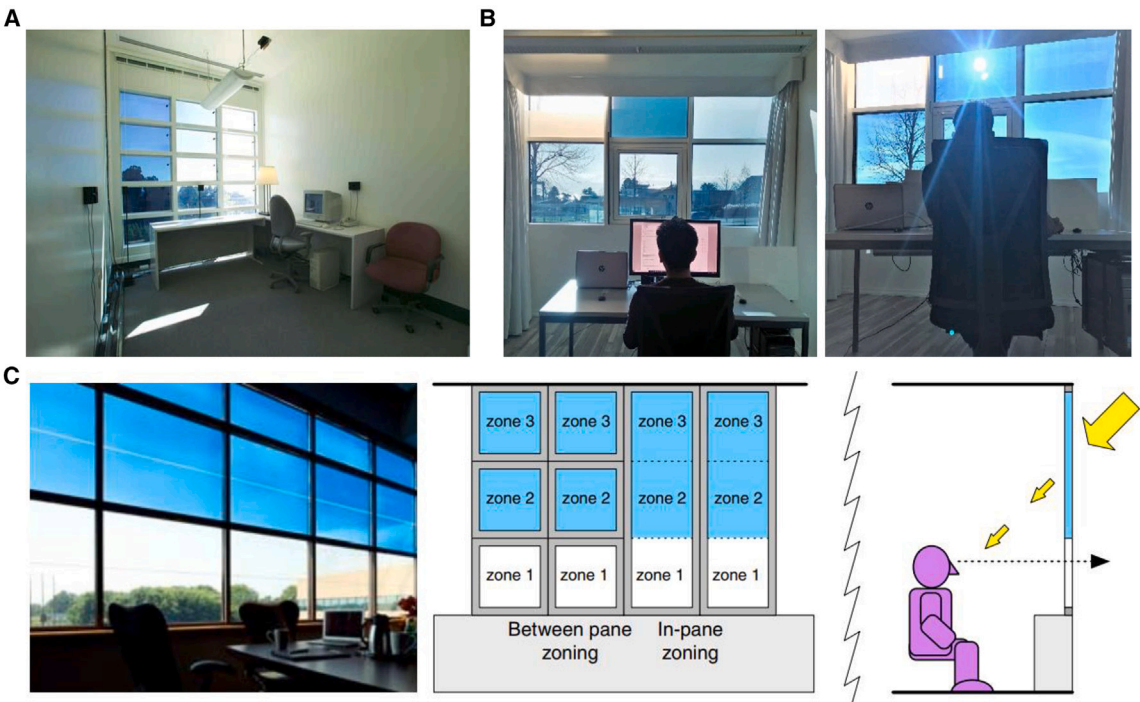


Figure 5. Experiments with EC glass
(A) Offices with EC glass and blinds. Each of the three window columns had an operable Venetian blind.⁹⁸
(B) Assessing user perception of glare from the sun in a controlled environment.¹⁰⁴
(C) Zoning of EC glazing by individual panes or within-pane zoning.¹⁰⁵
Reproduced with permission from (A) Clear et al.,⁹⁸ Elsevier; (B) Jain et al.,¹⁰⁴ Elsevier; (C) Mardaljevic et al.,¹⁰⁵ John Wiley and Sons.

The control strategies of ECWs are the focus of building simulation studies.^{106–108} The most convenient way is to assume that the glass is colored at a specific time, so it can be given different optical properties directly by modifying the schedule corresponding to the state of the glass.¹⁰⁹ The EC state can also be dynamically controlled by setting thresholds for other environmental parameters, such as indoor light levels, solar radiation, or indoor temperature.¹⁰⁶ This control logic can be achieved by setting up sensors and actuators with the energy management system (EMS) function of EnergyPlus.¹¹⁰ It is also possible to alter the EC glass state during simulation by transient system simulation tool (TRNSYS) reading files.¹¹¹ The simulation explorations conducted by different software packages are summarized in Table 5.

Based on specific control strategies, investigators have discussed the applicability of ECWs in different climate zones,^{109,112} compared the influence of internal disturbance and schedule differences between residential and office buildings for ECW application,¹¹⁰ and analyzed the optimal orientation and WWR of ECWs.^{113,114} Because several interacting parameters affect the energy performance of buildings, it is necessary to use parametric approaches for building simulation. When discussing the performance of ECWs, appropriate solutions should be given for case-specific requirements and not generalized.

PERFORMANCE REGULATION OF SMART WINDOWS

Performance of the ideal smart windows

In buildings, the performance of windows is usually measured by a shading coefficient (Sc) and heat transfer coefficient (K). To better distinguish between visible and NIR radiation, T_{lum} and Sc are replaced by an index, the light-to-solar gain ratio (LSG), which is also called luminous efficacy (Ke) in other literature.⁷³ The LSG is defined as the ratio of T_{lum} to solar heat gain coefficient (SHGC),¹²¹ which is a measure of the glazing's efficiency in penetrating visible solar radiation while blocking infrared light or heat. The SHGC is also known as total solar energy transmittance or g value. Theoretically, visible light and NIR light are believed to account for 47% and 50% of the solar radiation reaching the sea level under sunny conditions, respectively, so the maximum LSG is 2.13, but this value cannot be reached in practice. The LSG has a minimum value of 0.77 when T_{lum} is 0.6, and it is 1.03 when T_{lum} and NIR transmittance (T_{NIR}) are equal. K should be as small as possible and meet the thermal requirements of specific climate zones to ensure thermal insulation performance. The performance of the window can be comprehensively assessed by employing two parameters, K and LSG, which represent the performance of thermal insulation and the performance of light transmission and shading of the window, respectively.

Figure 6A shows a performance comparison of ordinary windows, LowE windows, and smart windows in winter and summer. The performance of ordinary windows and LowE windows is statically constant throughout the year, where ordinary windows have high absorption (α_{LWIR}) and emissivity (ϵ_{LWIR}) of long-wave infrared (LWIR), and LowE windows have low LWIR emissivity (ϵ_{LWIR}) and high LWIR reflectance (ρ_{LWIR}), so the thermal insulation is better than that of ordinary windows. LowE windows achieve a reduction in K by controlling radiant heat gain, but the Sc is small throughout the year, leading to a limited passive heating capacity in winter. The Sc of the glass is not anticipated to remain constant for zones with distinct winter and summer throughout the year.¹²² The performance of smart windows is changing dynamically according to seasonal demands. It is desired that in summer, the solar radiation transmittance (T_{sol}) and Sc are smaller, whereas the

Table 5. Several building simulation investigations associated with ECWs

Reference	Simulation software	Control strategy	Location	Building type	Building size	WWR	Window orientation	$T_{lum,b};$ $T_{lum,c}$	$T_{sol,b};$ $T_{sol,c}$	$K_{b};$ K_c	Concerns	Specific indicators
Wang et al. ¹⁰⁹	Ecotect	temperature	Zhuhai	office	533 m ³	N/A	all surfaces	0.76; 0.12	0.74; 0.20	2.1; 2.1	energy; daylight	monthly energy consumption; average illumination
Cannavale et al. ¹⁰⁶	EnergyPlus	illuminance; glare; irradiance	Rome	office	70 m ³	42%	south	0.409; 0.027	0.439; 0.113	1.98; 1.98	energy; daylight	energy consumption; useful daylight illuminance (UDI), daylight glare index (DGI)
DeForest et al. ¹¹⁰	EnergyPlus; COMFEN	energy consumption	16 cities in the United States	office; residential	N/A	35%	south	0.780; 0.667	0.646; 0.297	1.804; 1.625	energy	internal heat gains
Kheybari et al. ¹¹¹	radiance, TRNSYS	model, predictive control	Mannheim	office	99 m ³	85%	south	0.44, 0.007	0.29, 0.004	1.3, 1.3	energy, daylight, thermal comfort	energy consumption, daylight glare probability (DGP), UDI, predicted mean vote (PMV)
Tavares et al. ¹¹²	ESP-r	temperature; irradiance	Coimbra	office	196 m ³	20%	variable	0.62; 0.035	0.48; 0.090	1.6; 1.6	energy	energy needs
Lahmar et al. ¹¹³	Grasshopper; Rhinceros	illuminance	Biskra	residential	57 m ³	variable	southwest	0.64; 0.01	0.44; 0.10	1.9; 1.9	energy	energy consumption
Xing et al. ¹¹⁴	EnergyPlus	shading	Shanghai; San Francisco	office	90 m ³	variable	variable	0.666; 0.021	0.445; 0.033	2.494; 2.494	energy	annual energy consumption; energy-saving percentage
Favoino et al. ¹⁰⁸	EnergyPlus; Daysim	rule-based control strategies	Sydney; Rome; London	office	112 m ³	60%	south	0.595; 0.238	0.508; 0.238	2.0; 2.0	energy; daylight	the total specific yearly site energy; net total specific yearly site energy; load profile matching index; grid interaction index; UDI; DGP
Lee and Tavi ¹¹⁵	DOE-2	irradiance; illuminance	Houston; Chicago	office	38 m ³	30%; 60%	south	0.45; 0.02	0.37; 0.10	2.76; 2.76	energy; daylight	annual energy use; peak electric demand; average annual daylight illuminance; DGI
Fernandes et al. ¹¹⁶	Radiance	visual comfort and illuminance	Oakland	office	38 m ³	N/A	south	0.60; 0.05	N/A	N/A	energy; daylight	lighting energy; average illumination
Kim and Han ¹¹⁷	Grasshopper	illuminance	Gwangju	office	450 m ³	N/A	south	0.7; 0.2	N/A	N/A	daylight	LEED v.4.1 evaluation
Bui et al. ¹¹⁸	EnergyPlus; Python	irradiance	Melbourne; Texas	office	19,980 m ³	60%	four cardinal directions	0.728; 0.014	0.522; 0.187	3.534; 3.534	energy	purchase cost; energy consumption; payback period
Paule et al. ¹¹⁹	DIAL+	illuminance; glare	Lausanne	office	408 m ³	40%	west	0.59; 0.01	0.4; 0.05	1.1; 1.1	energy; daylight	average illuminance; indoor temperature
Park et al. ⁹¹	eQUEST	schedule	Seoul	office	19,000 m ³	N/A	south	0.66; 0.10	0.52; 0.12	1.27; 1.27	energy	energy cost; annual and peak day energy
Dussault and Gosselin ¹²⁰	Daysim; Radiance; TRNSYS	irradiance; illuminance; window heat flux	10 cities in the United States and Canada	office	300 m ³	33%; 50%; 67%	variable	0.621; 0.015	0.381; 0.010	1.63; 1.63	N/A	N/A

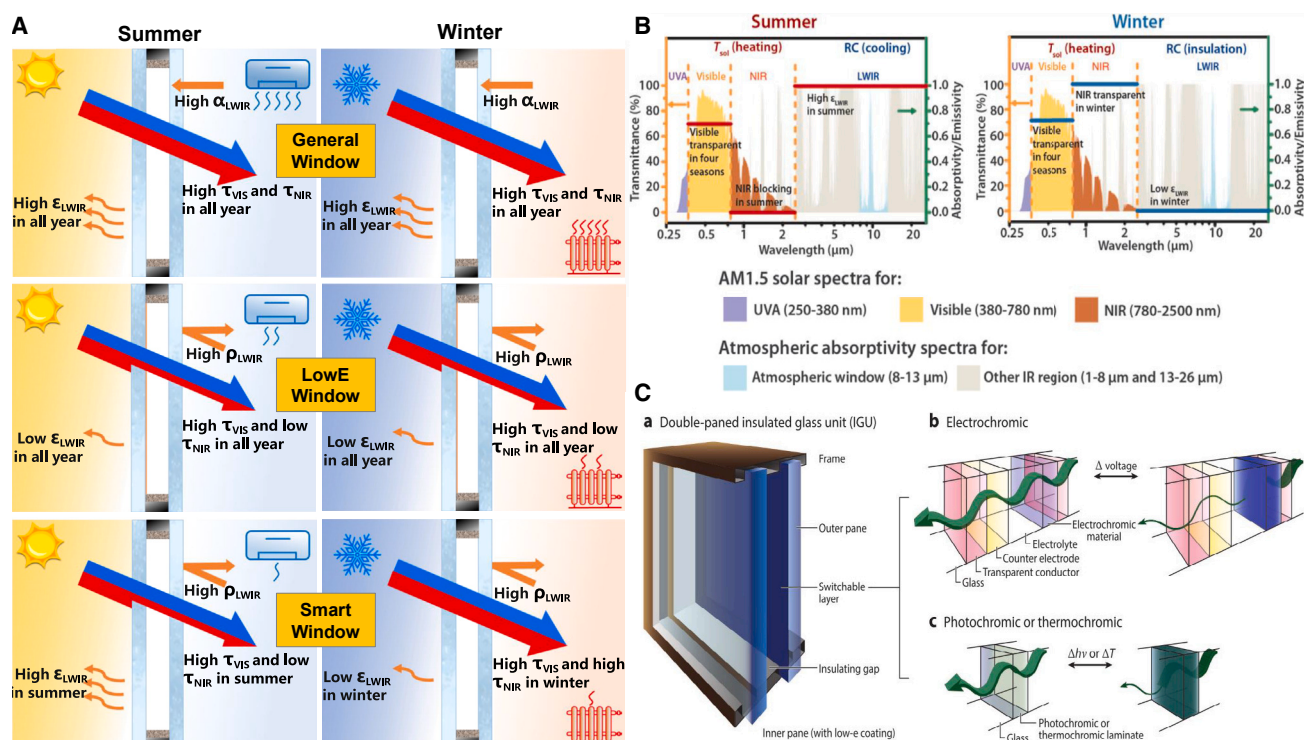


Figure 6. Performance and structure of an ideal smart window

(A) Performance comparison of an ordinary window, LowE window, and smart window.

(B) Optical properties of the ideal smart window at various wavelengths.³⁹

(C) Insulated glass unit combined with LowE glass and switchable layer-coated glass.⁸⁴

Reproduced with permission from (B) Wang et al.,³⁹ The American Association for the Advancement of Science; (C) Wang et al.,⁸⁴ Annual Reviews.

LSG and K are larger, reducing heat gain while promoting heat dissipation. In addition, higher T_{sol} and S_c and lower LSG and K are desired in winter to enhance heat gain and reduce heat loss. The inner surface of the glass should reflect the long-wave radiation emitted from indoor objects (cold radiation in summer and heat radiation in winter) back into the room to preserve the room temperature.

Figure 6B further demonstrates the modulation performance of smart windows in the visible, NIR, and LWIR bands. The ideal smart window should consistently maintain 60% or higher T_{lum} throughout the year, automatically shield NIR light during intense solar radiation, and maintain high ϵ_{LWIR} for increased radiative cooling in summer, particularly in the 8- to 13- μm atmospheric window. In winter, the ideal smart window should have a high T_{NIR} while maintaining a low ϵ_{LWIR} to suppress radiative cooling.^{39,123}

In building applications, the smart window structure can be represented as an insulated glass unit combined with LowE glass and switchable layer-coated glass.⁸⁴ The switchable chromogenic coating is located on the inner surface of the external glass panel, and the LowE coating is placed in the inner pane, as shown schematically in Figure 6C.

Performance regulation of TCWs and ECWs

In the present work, we depict the performance distribution of different windows by K and LSG, as demonstrated in Figure 7. The performance data of TCWs and ECWs were obtained from the literature and listed smart window commercial products. The arrows in the figure indicate the darkest state pointing to the clearest state; that is, TCWs from the high-temperature state to the low-temperature state

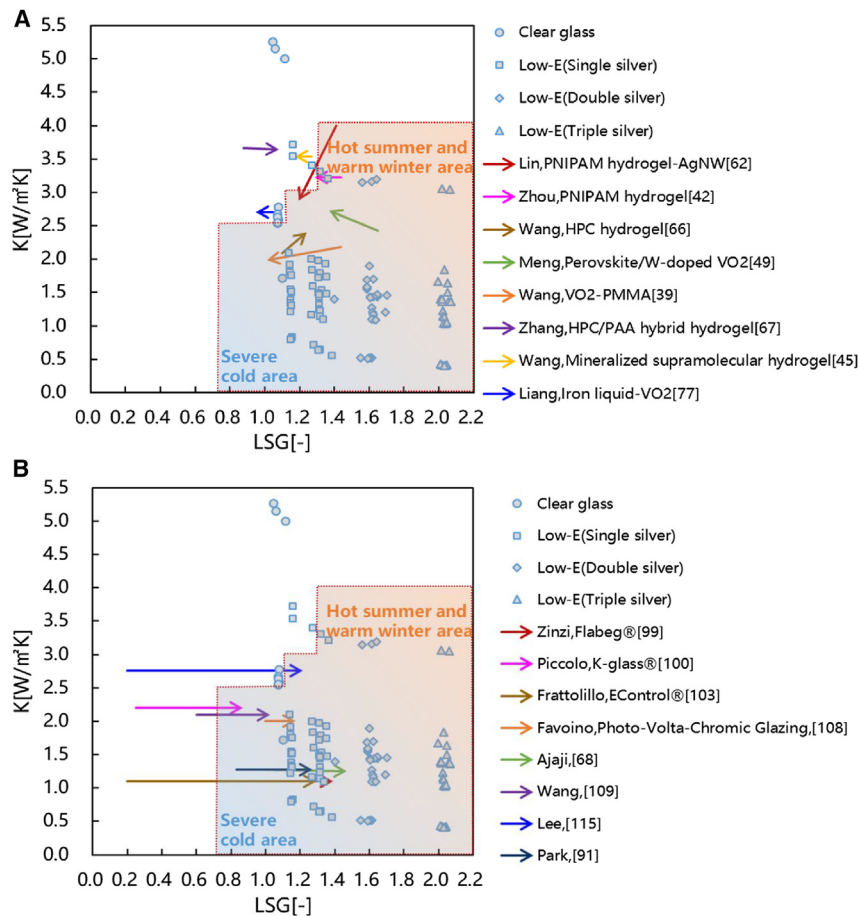


Figure 7. Applicability distribution map of existing smart window technologies

(A) Performance distribution of TCWs as façade windows.

(B) Performance distribution of ECWs as façade windows.

and ECWs from the colored state to the transparent state. The energy savings and applicability of windows are closely associated with the climatic region. For thermal zoning in China, the standard specifies the threshold values of K and SHGC in each climate zone, as displayed in Table 6. Building applications should ensure that windows have a visible light transmittance of at least 0.6; therefore, the LSG range for windows in each climate zone was calculated based on the SHGC limit values. Using the ranges of K and LSG as boundary lines, the colored areas of Figure 7 give the range of performance requirements for façade windows, whereas Figures S3 and S4 provide the range for skylights. The arrows for TCWs and ECWs reflect the performance distribution of different types of windows, as well as the applicability of existing technologies in different regions.

In Figure 7, the LSG of clear glass ranges from 1–1.1; LowE glass shields part of the NIR, so the LSG is between 1.1 and 2.1. The LSG of TCWs and ECWs can be adjusted with the state, but the interval between the clearest and darkest state is narrow for most products,²⁵ so the regulation performance is limited. The arrows in the K -LSG diagrams indicate that the performance of smart windows in hot summer points to that in cold winter, so ideally it should point from the larger K and LSG to the smaller K and LSG; that is, from the upper right to the lower left. The performance

Table 6. The parameters of climate zoning and windows in China

Climatic region	Zoning rules	Threshold value of K ($W/[m^2K]$) ¹²⁴	Threshold value of the SHGC		Threshold value of the LSG ($T_{lum} = 0.6$)	
			Façade window	Skylight	Façade window	Skylight
Severe cold area	$3,800 \leq HDD18^a$	≤ 2.0	no limit	no limit	≥ 0.77	≥ 0.77
Cold area	$2,000 \leq HDD18 < 3,800$	≤ 2.5	no limit	≤ 0.44	≥ 0.77	≥ 1.4
Hot summer and cold winter area	$700 \leq HDD18 < 2,000$	≤ 3.0	≤ 0.52	≤ 0.35	≥ 1.15	≥ 1.7
Hot summer and warm winter area	$HDD18 < 700$	≤ 4.0	≤ 0.48	≤ 0.30	≥ 1.25	≥ 2.0

^aThe heating degree days (HDD) are calculated as the sum of the discrepancies between daily average temperatures and the base temperature ($18^\circ C$).

of several studies in Figure 7A fits this pattern but also has the opposite arrows. Wang et al.⁶⁶ obtained a variable ϵ_{LWIR} by flipping the window. The K values of the tunable emissivity TCWs were reported to be 2.09 and 2.4 in summer and winter, respectively. It has the configuration of low outer-surface ϵ_{LWIR} and high inner-surface ϵ_{LWIR} in winter, slightly drifting from the ideal window (low outer-surface ϵ_{LWIR} and low inner-surface ϵ_{LWIR} in winter). But it proved to be more energy efficient than LowE windows. The arrows in Figure 7B horizontally point from small LSG to large LSG because the T_{lum} of most ECWs in the tinted state is too small to avoid glare, but also at the expense of good lighting. For the boundary line divided by T_{lum} of 0.6, the performance of some ECWs in the tinted state is outside of the colored region and does not meet the requirements of practical applications. The LSG span of ECWs is larger, and that of TCWs is smaller. In addition, most ECW studies believed that the K value of the window is constant throughout the year, but a few TCW studies emphasized the possibility of regulating K in different seasons by adjusting emissivity, which is reflected by slanted arrows in Figure 7A.

Figure 7 also reflects some of the shortcomings of the existing technologies. Some of the arrows in Figures 7A and 7B are rarely completely in the coloring area enclosed by the boundary lines; some of the T_{lum} in the dark state is too small, and some of the K is too large to meet the practical application requirements. Therefore, for future development of TCWs and ECWs, it is necessary to first satisfy the reasonable T_{lum} for all seasons and achieve better thermal insulation performance through a reliable structure and second to improve the shading-adjustable and heat transfer-adjustable methods based on large T_{lum} and small K .

DIFFERENCES BETWEEN TCWs AND ECWs

TCWs and ECWs have different core logic, and the technologies of TCWs and ECWs also show different arrow trends in Figure 7. TCWs passively regulate solar radiation, and the main factor affecting their application is climate conditions, while the application of ECWs depends on their color-changing control strategies. Therefore, we discuss their applicability differences.

Climate adaptability of TCWs

The energy-saving potential and daylight regulation ability of windows depend extensively on climatic conditions.¹²⁵ In Figure 7, windows with performance

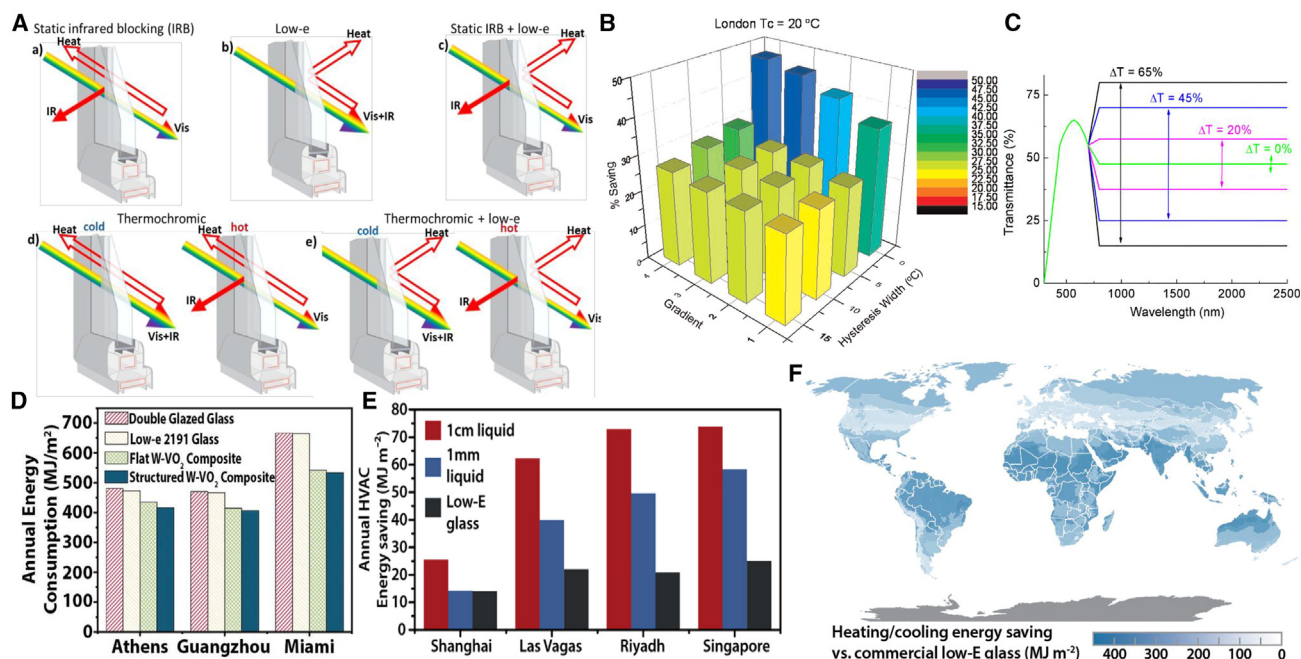


Figure 8. Exploration of climate adaptation of TCWs

(A) Different combinations of static glass and dynamic glass.¹²⁶

(B) Effect of the thermochromic transition temperature, gradient, and hysteresis width on energy savings.¹³⁰

(C) Ideal thermochromic spectra for NIR modulation.¹³¹

(D) Annual energy efficiency of W-VO₂-based smart windows in cities near the Tropic of Cancer.⁷⁸

(E) Annual energy efficiency of PNIPAM hydrogel-laminated smart windows in different cities.⁴²

(F) Energy savings potential of RCRT windows in all regions of the world.³⁹

Reproduced with permission from (A) Mann et al.,¹²⁶ MDPI; (B) Warwick et al.,¹³⁰ Elsevier; (C) Saeli et al.,¹³¹ Elsevier; (D) Zhou et al.,⁷⁸ John Wiley and Sons; (E) Zhou et al.,⁴² Elsevier; (F) Wang et al.,³⁹ The American Association for the Advancement of Science.

distribution in the lower left corner of the colored area are more suitable for colder climates, while windows in the upper right corner are more appropriate for warmer climates. In warmer areas, larger K and LSG facilitate heat dissipation and shade, respectively. For static glass, NIR-blocking glass is suitable for hot areas, and high-transmittance LowE glass is suitable for cooler areas.¹²⁶ For dynamic glass, TCWs stimulate the thermochromic process through a temperature phase transition, which mainly realizes energy savings by reducing the cooling demand of the building. Therefore, cold regions dominated by heating consumption will not have suitable climatic conditions for TCWs.^{127,128} For thermochromic materials subjected to high t_c , the excitation of the phase change process can be accelerated by auxiliary heating.^{64,129} Composite windows combining TCWs with LowE glass of various transmittance can also be selected to meet the needs of K and LSG, depending on the climate conditions (Figure 8A).

There are two ways to explore the suitability of smart window materials; one is to prepare material samples and verify their application performance through experiments or simulations. Zhou et al.⁷⁸ simulated the annual energy consumption of W-VO₂ TCWs applied in cities near the Tropic of Cancer. The results showed that significant energy savings were achieved in all cities, and the phenomenon was more pronounced in Miami (Figure 8D). They also conducted outdoor small-room experiments using PNIPAM hydrogel glass samples in Beijing, Guangzhou, and Singapore and single-story office building simulations in Shanghai, Las Vegas, Riyadh, and Singapore. Compared with ordinary glass windows, 1-cm hydrogel-laminated smart

windows can reduce HVAC energy consumption by 19.1%, 24.3%, 25.4%, and 44.6% per year in the four cities mentioned above, respectively (Figure 8E).⁴² Wang et al.³⁹ conducted energy consumption simulations using a typical 12-story office building model and showed that passive radiative cooling-regulating thermochromic (RCRT) windows produced higher energy savings in all different climate zones, with energy savings up to 324.6 MJ m^{-2} (Figure 8F).

The other is to make requirements for material performance based on the climatic characteristics of specific regions, which is often achieved by building simulations of the ideal spectrum. Liang et al.¹²⁷ simulated the energy and daylight performance of VO₂ TCWs applied to a typical office, showing that a low transition temperature (i.e., 20°C) is not necessary for building energy efficiency and that a larger solar transmittance modulation is preferable for most climatic conditions. Mann et al.¹²⁶ demonstrated through simulations that, for intermediate climates, the combination of adaptive thermochromic glass with LowE coatings is well suited and that this glazing system can lead to annual cost savings of €638 per dwelling (172 m², 25% window facade) and annual nationwide CO₂ savings of 4.5 Mt. Saeli et al.¹³¹ showed, by ideal spectral simulation, that large changes in optical properties in the NIR and low transition temperatures can optimize energy-saving performance (Figure 8C). Warwick et al.⁷⁴ derived from spectral simulations of ideal TCWs that the transition temperature has a much greater effect on the energy demand reduction properties of the glass compared with the hysteresis gradient. The general rule is reflected in hot, cold, or mixed climate zones; lower transition temperatures and steeper transition hysteresis gradients lead to greater energy savings. It is further noted that the best parameter for reducing energy demand is the combination of narrow and sharp hysteresis behavior with low transition temperatures (Figure 8B).¹³⁰

Control strategy of ECWs

The design of control strategies plays a fundamental role in the performance of ECWs. Control strategies are closely related to energy efficiency, and different target conditions may lead to completely dissimilar results. Most control strategies for switchable windows in the literature are based on simple rule-based control, such as illuminance, temperature, glare, and irradiance. Research work at Lawrence Berkeley National Laboratory (LBNL)¹³² showed that daylight-based control algorithms produce the best overall annual energy performance. Cannavale¹⁰⁶ compared the effects of three control strategies (illuminance-based, glare-based, and irradiance-based) on energy savings and day lighting of ECWs, and the results revealed that the illuminance-based control strategy is the ideal compromise between maximum year-round energy savings and day lighting utilization. Platzer¹³³ found that switching based on room temperature produced the lowest initial energy consumption. Tavares et al.¹¹² investigated the influence of different environmental factors on the energy supply of ECWs and proved that adjusting the optical properties of ECWs based on incident radiation gives the best results in a Mediterranean climate.¹³⁴ The inconsistency of these results revealed the need for continued argumentation in the field. Moreover, many studies have mainly investigated the impact of control strategies on the energy performance of smart windows, with little analysis of the impact on visual comfort and perception. A well-designed control strategy should be able to fully consider the comprehensive performance of smart windows to achieve energy-saving performance while realizing improvement of the indoor environment and personnel comfort.¹¹¹

In some climates or specific use scenarios, a single control strategy based only on solar radiation or glare may lead to higher energy consumption and lower visual

comfort. A few studies have been devoted to more advanced control strategies, such as a proportional-integrative-derivative control based on work-plane illuminance and an adaptive neuro-fuzzy inference control system. Favoino et al.¹⁰⁸ evaluated various control strategies, from passive and simple rule-based controls to optimized rule-based and predictive controls, showing that more advanced predictive control strategies can achieve a balance of energy savings and comfort. The control of ECWs should be integrated with dimmable artificial lighting¹⁰⁵ to deliver optimal lighting levels during use. In addition, the applicability of smart windows should be differentiated according to the type of office or residence. Peak heat gain in office buildings usually occurs between 9 a.m. and noon, while it occurs between 6 p.m. and 8 p.m. in residential buildings.¹¹⁰ The state switching of the windows should correspond to the building load characteristics to minimize energy demand.

To achieve a good control strategy, ECWs should improve our vision of system integration.¹³⁵ ECWs must have a complete power network, and the development of solar cells,¹³⁶ crystalline silicon-based photovoltaic (PV) modules,¹³⁷ and translucent PV windows¹³⁸ offers a strong guarantee for application of ECWs. The power generation capacity of solar cells and the electricity demand of ECWs are complementary because self-powered tinted windows allow sufficient electrical energy to be supplied when the sun is strong, forcing the window to quickly change from transparent to dark tinting.¹³⁹ Meng et al.⁴⁹ prepared a smart PV window foil with NIR modulation and low long-wavelength IR emissivity. The values of peak power conversion efficiency were reported to be 15.4% and 16.1% at 25°C and 45°C, respectively, and the average visible light transmission was 25.5%. Ma et al.¹⁴⁰ proposed a smart window consisting of a building-integrated PV (BIPV) panel and an all-solid-state EC stack. The variable output voltage from the BIPV module is employed to drive the EC stack and achieve interior lighting modulation. PV cells can also be combined with thermochromic layers. Liu et al.¹⁴¹ developed a BIPV smart window system based on a hydroxypropyl cellulose (HPC) thermochromic hydrogel film, which realized the dual functions of self-generation and adaptive solar radiation control.

While integrated translucent PV windows can potentially enhance energy efficiency, the presence of solar cells can lead to undesired shadows and result in low-quality daylight.¹⁴² The two chief goals of solar cell window optimization are to improve day lighting and power generation.¹⁴³ These remain the focus of continued future research. The combination of energy storage and adaptive power generation should also be evaluated^{31,144} to enhance the flexibility of the control strategy.

CONSISTENCIES BETWEEN TCWs AND ECWs

As new technologies for dynamic envelopes, the common goal of TCWs and ECWs is to enhance and optimize their regulation performance to keep approaching the ideal smart window. In practical application, both technologies should pay more attention to comprehensive evaluation of energy-saving performance, the built environment, and human comfort. In this section, we discuss in detail the consistencies of these two techniques.

Improvement of dynamic regulation performance

The LSG and K of an ideal smart window should be regulated independently, allowing it to meet the demand at any time of the year. As shown in Figure 9, for light transmission or shading performance, high T_{lum} should be guaranteed first, and LSG can be adjusted by varying the T_{sol} of the window between 0.25 and 2.5 μm , with corresponding measures including thermochromic and EC coatings, blinds,

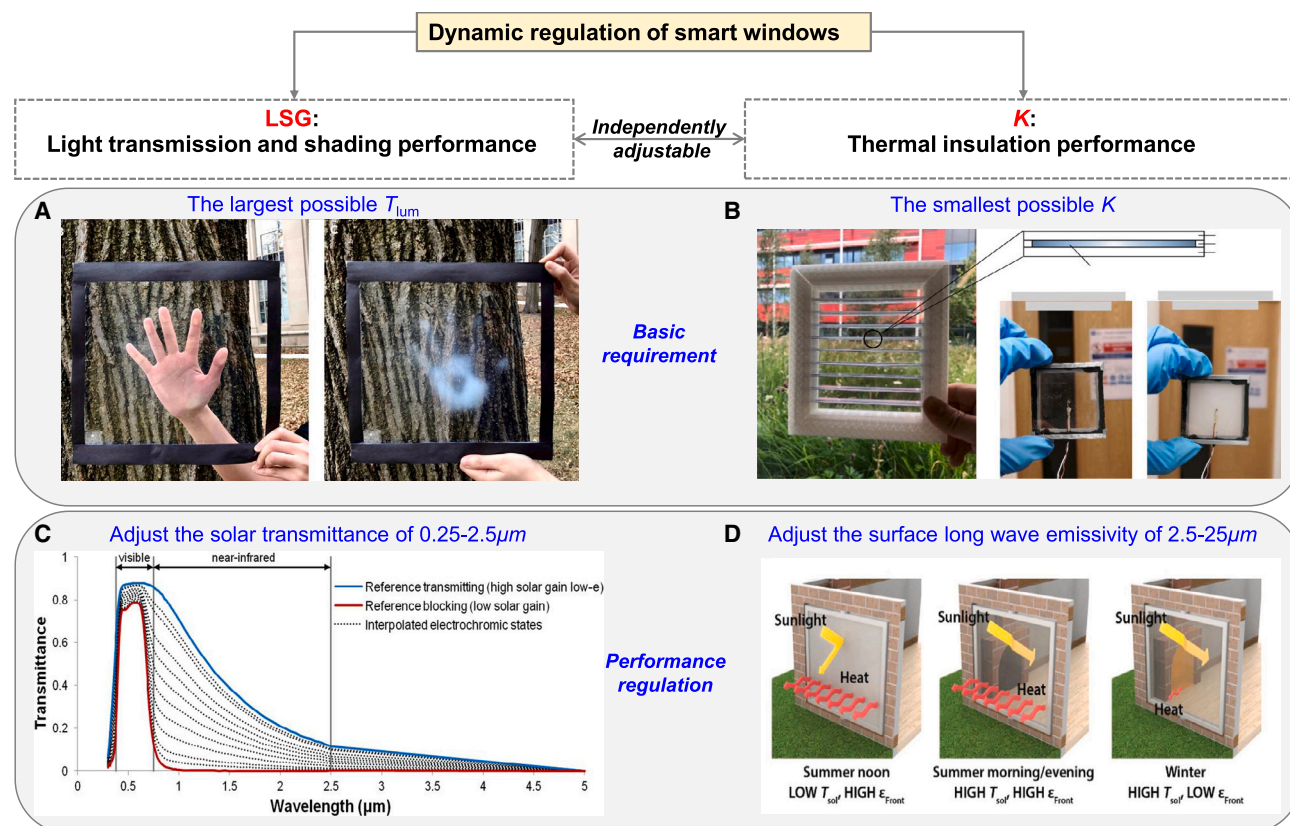


Figure 9. The dynamic regulation of smart windows

(A) High T_{lum} windows based on PNIPAM-AEMA microparticles.⁶⁰

(B) Parallel slat structure formed by the TIM and thermochromic material.¹⁴⁵

(C) Schematic spectra of NEC glass.¹¹⁰

(D) Emissivity adjustment by flipping the window panel.⁶⁶

Reproduced with permission from (A) Li et al.,⁶⁰ Elsevier; (B) Sun et al.,¹⁴⁵ Elsevier; (C) DeForest et al.,¹¹⁰ Elsevier; (D) Wang et al.,⁶⁶ Elsevier.

and rolling curtains. For thermal insulation performance, first a low K value is maintained by filling gas and multi-layer window structure, and then K can be regulated by variation of surface emissivity in the long-wave radiation band from 2.5–25 μm . The relationship between surface emissivity and K is shown in Equations S1 and S2 in Note S2. Therefore, the dynamic regulation performance of smart windows can be briefly stated as the largest possible T_{lum} , the smallest possible K , and independently adjustable values of K and LSG.

Few studies have emphasized the independent tunability of K and LSG. Several ways for setting K and LSG are described in the following for reference.

For the LSG, Li et al.⁶⁰ reported a hydrogel smart window based on PNIPAM-AEMA microparticles with high transmittance of 87.2% at 25°C, breaking the technical bottleneck of high T_{lum} (Figure 9A). DeForest et al.¹¹⁰ proposed NIR switching EC (NEC) glass with the spectrum shown in Figure 9C, which can achieve high visible light transmission while maintaining NIR modulation. Independent or selective modulation of visible and NIR bands is an effective method to modulate LSG, which can provide better compatibility in different regions. This can be achieved by solution-processed nanocomposites⁸⁴ or by a combination of two different functional materials.⁷⁷

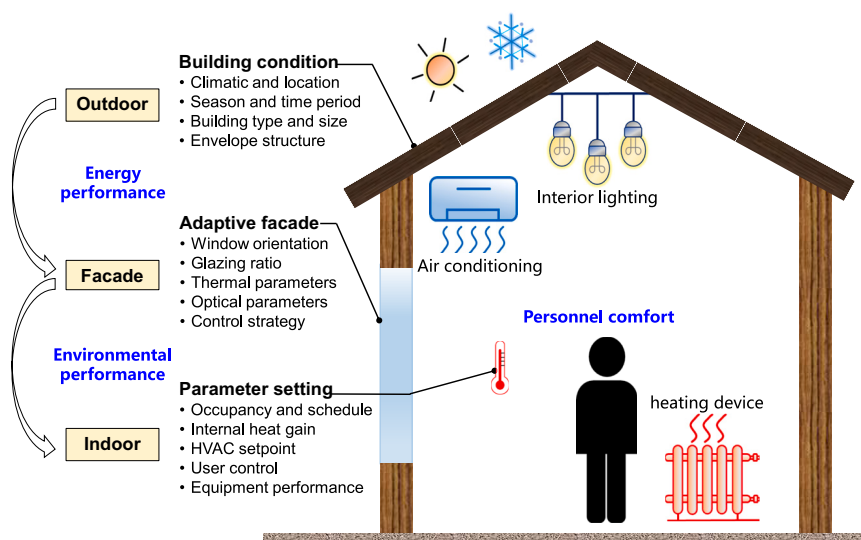


Figure 10. Comprehensive evaluation of energy, environment, and comfort

Characteristics of the ideal adaptive smart window: minimizes energy consumption; adjusts its thermal and light characteristics to improve the quality of the indoor environment; meets the comfort of the occupants.

For K, Sun et al.^{145,146} reported a multi-effect smart window integrating thermochromic materials and transparent insulation material (TIM). It is capable of lessening heat loss by increasing thermal resistance while efficiently transmitting solar energy to make a daylight environment¹⁴⁷ (Figure 9B). Zhao et al.¹⁴⁸ embedded VO₂ nanoparticles in an ultralow thermal conductivity aerogel film, and this 3.0-mm-thick film could obtain a low U value of $\sim 3.0 \text{ W}/(\text{m}^2\text{K})$ as well as a high light transmission of more than 60% and ΔT_{sol} of $\sim 20\%$. Lin et al.⁶² used a temperature-triggered water capture-and-release mechanism caused by the PNIPAM phase transition to enable windows with different emissivity in winter and summer, achieving simultaneous regulation of solar energy transmission and thermal radiation. Ye et al.¹⁴⁹ stated that the energy-saving potential of adjusting window emissivity is greater than that of adjusting solar transmittance. Wang et al.³⁹ prepared TCWs by dispersing VO₂ nanoparticles in poly(methyl methacrylate) (PMMA) solution, providing different emissivity at high (ϵ_{LWIR} of 0.61) and low (ϵ_{LWIR} of 0.21) temperatures to automatically control radiative cooling. Long et al.⁶⁶ achieved high emissivity in summer (ϵ_{LWIR} of 0.95) and LowE in winter (ϵ_{LWIR} of 0.1) by flipping the window panels (Figure 9D). The application of radiative sky cooling in the building sector shows great and growing potential.¹⁵⁰ Zhou et al.¹⁵¹ established a radiative cooling transparent film for the daytime cooling of smart windows, which can be exploited as a rolling curtain to produce a remarkable impact on indoor cooling. A combination of roof daytime passive radiative coolers and facade TCWs can provide 17% energy savings in Hong Kong.¹⁵²

Comprehensive evaluation of energy, environment, and comfort

The generalization of tests and simulations reveals that smart window technologies hold exciting promise for building applications. As can be seen from the concerns and indicators listed in Tables 3 and 5, researchers have explored smart windows mainly to verify the feasibility of material technologies through energy performance, with some neglect of the physical built environment and human comfort. In fact, as a connection between indoor and outdoor environments, the window contributes equally to energy performance and environmental performance,¹⁵³ and the latter directly affects personnel comfort. Figure 10 demonstrates the comprehensive evaluation concept based on

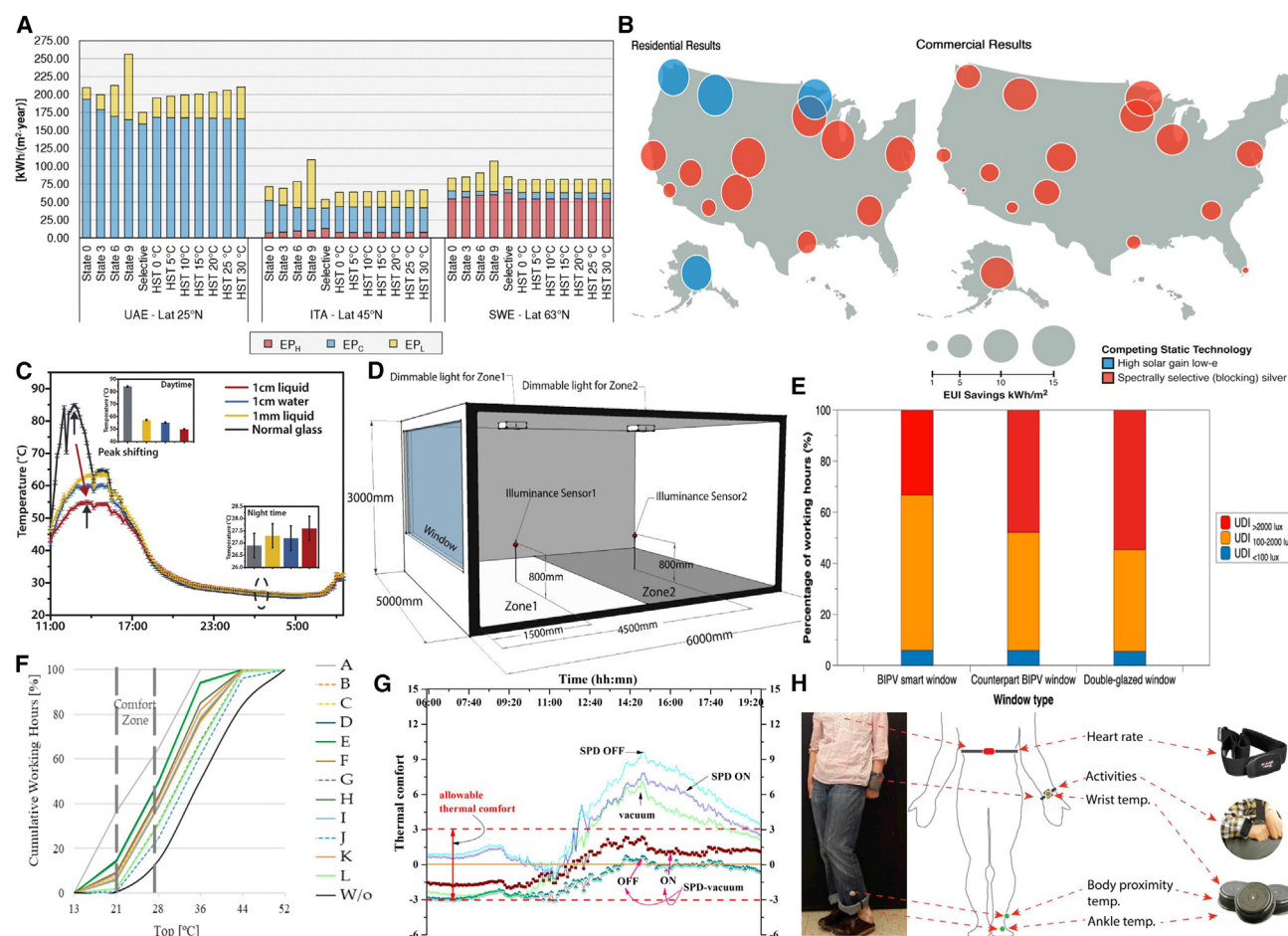


Figure 11. Representative studies on energy, environment, and comfort

(A) Energy-saving performance of LETC for applications in different climatic regions.¹⁵⁴

(B) Energy-saving performance of NEC glass for United States application.¹¹⁰

(C) The role of hydrogel smart windows in shifting peaks.⁴²

(D) Daylight performance test for a typical office.¹²⁷

(E) Comparison of lighting performance between BIPV smart windows and traditional static windows.⁶⁹

(F) Cumulative annual working time comfort percentage for applying different types of SCFs on glazing systems.¹⁵⁵

(G) Thermal comfort analysis of SPD-vacuum glazing.⁷²

(H) Semi-invasive physiological sensors and wearing locations.¹⁵⁶

Reproduced with permission from (A) Giovannini et al.,¹⁵⁴ Elsevier; (B) DeForest et al.,¹¹⁰ Elsevier; (C) Zhou et al.,⁴² Elsevier; (D) Liang et al.,¹²⁷ Elsevier; (E) Liu et al.,⁶⁹ Elsevier; (F) Teixeira et al.,¹⁵⁵ Elsevier; (G) Nundy et al.,⁷² Elsevier; (H) Liu et al.,¹⁵⁶ Elsevier.

the heat transfer path. The factors influencing comprehensive performance are divided into three categories: outdoor building conditions, building adaptive façade, and indoor parameter settings. As shown in Figure 10, an ideal adaptive façade can be defined as a façade that minimizes the total energy use of an interior space while adjusting its thermal and light characteristics to improve the level of indoor environmental quality to meet the comfort of the occupants.⁷⁵ Whether thermochromic or ECWs, comprehensive evaluation of energy performance, environmental performance, and personnel comfort deserves to be focused on, which provides a new and important perspective for assessing the performance of smart windows.

Figure 11 presents representative studies on energy, environment, and comfort evaluation. In terms of energy performance, Giovannini et al.¹⁵⁴ adopted the

simulation method to evaluate the itemized energy consumption and visual comfort in an office space integrating Ligand exchange thermochromic glazing (LETC) in different climatic zones. The results showed that LETC glass achieves a 3%–10% reduction in total energy use and daylight availability improvements from 5% to nearly 20% compared with a static glass benchmark with comparable selectivity (Figure 11A). DeForest et al.¹¹⁰ simulated the energy performance of NEC glass used in 16 cities in the United States. Commercial heating and cooling savings range from 5–11 kWh/m² per year, and residential savings range from 8–15 kWh/m² per year, in many regions on the order of 10% (Figure 11B).

In terms of light and thermal environmental performance, the outdoor experiments by Zhou et al.⁴² showed that hydrogel TCWs could delay the peak temperature by 2 h at hot noon, which would facilitate transfer of electricity to lower electricity prices (Figure 11C). Simulations by Liang et al.¹²⁷ demonstrated that TCWs provide better light performance than clear double glazing (useful daylight illuminance [UDI]_{500–2,000 lux} increase of up to 15.52%) (Figure 11D). Liu and Wu⁶⁹ compared the day lighting performance of BIPV smart windows with traditional static windows, showing that BIPV smart windows can significantly increase the percentage of UDI_{100–2,000 lux} to 60.9% while reducing the risk of overlighting (Figure 11E).

In terms of comfort, Teixeira et al.¹⁵⁵ evaluated the energy and environmental performance of different types of solar control films (SCFs) in glass systems and also quantified the comfort of SCFs by the percentage of comfort during working hours (Figure 11F). Nundy et al.⁷² used predicted mean vote (PMV) and Predicted Percentage of Dissatisfied (PPD) to analyze the thermal comfort of switchable suspended particle device (SPD)-vacuum glazing (Figure 10G). The quality of the indoor environment affects the comfort of people and impacts productivity. Therefore, field evaluation tests with subject participation are extremely important. Non-invasive data collection is highly beneficial for exploring the impact of occupant behavior on energy use (Figure 11H), where failure to consider occupant behavior can lead to increased uncertainty in energy performance.¹⁵⁷

In summary, in the gradual transition of smart window technology to building applications, attention should be paid not only to its energy performance but also to the built environment oriented by personnel comfort.

MATERIAL AND BUILDING REQUIREMENTS FOR TCWs AND ECWs

Material requirements

The ultimate goal of technological development is achieving large and reliable applications, so the exploration of materials cannot be decoupled from the needs of the building. The expected performance factors for smart windows in applications are shown in Table 7. In addition, essential indicators of structural stability, large-scale preparation, and acceptable cost should be guaranteed, and future materials development should make greater breakthroughs in these areas.

Existing technologies have been explored in depth for enhancing the stability of smart windows. Related technical enhancements were described in the previous section, and listed here are several representative studies. A durability-enhanced VO₂ nanoparticle-polymer thermochromic film was prepared by Zhao et al.⁷¹ The lifetime of VO₂ nanoparticles was greatly improved by a cross-linked polymer matrix with high molecular weight. This film was able to maintain 30% of its sunlight modulation capability after 900 h of accelerated durability experiments.⁷¹ PNIPAM particles

Table 7. The performance factors and essential indicators of smart windows for building applications

Performance factors of TCWs	Performance factors of ECWs
• $T_{lum} \geq 60\%$¹⁵⁸ • $\Delta T_{sol} \geq 50\%$ • t_c is close to the room temperature	• contrast ratio of at least 5:1 between maximum and minimum transmittance • application voltage in the range of 1–5 V • high CE¹⁵⁹
Essential indicators of TCWs and ECWs	
• switching time of several minutes • acceptable colors • sustained performance over 20–30 years¹⁵⁹ (25,000–50,000 cycles) • large-area production and acceptable cost	

were embedded in a water-rich silica-alumina-based gel matrix to prepare a flexible hybrid hydrogel that could be printed on a large area.⁵⁹ Hybrid hydrogels based on host-guest interactions can avoid shrinkage, resulting in highly reproducible and durable films.¹⁶⁰ Enhanced sealing may further increase the lifespan of thermochromic hydrogels.

Large-scale preparation techniques are closely related to the breadth of applications. The polymer-assisted deposition method enables VO₂ thin films with controlled morphology and thickness, and the low cost of chemical preparation and easy scale-up make it suitable for practical applications of the fabricated materials.³⁷ Nanomaterials can provide enhanced CE, faster switching, and longer life, and their compatibility with solution processing enables low-cost and high-throughput manufacturing.⁸⁴ The roll-to-roll production process makes it possible to produce flexible films continuously on a large scale, contributing to the manufacture of a new generation of high-quality smart windows.⁸⁰ Given the difficulty of large-scale TCW manufacturing, an artificial virtual thermochromic environment is an innovative solution.^{161–163}

Another important factor that hinders the application of smart windows is the cost. ECWs have been commercially available for decades but are expensive to apply, with an additional cost of about \$12/m² for ECWs compared with conventional insulating glass units; the payback period for residential buildings is about 30 years and for commercial buildings around 60 years,¹⁶⁴ which exceeds the design life of the building. Retrofitting rather than new construction may be the largest market for smart windows. In future development of smart windows, more natural organic polymer materials can be utilized to meet the green low-carbon criterion while reducing the cost of windows.¹²

Building requirements

Occupant well-being is directly related to a range of environmental factors, particularly daylight distribution, glare, and indoor thermal environments, which are all closely linked to windows.¹⁶⁵ Figure 10 emphasizes the importance of comprehensive evaluation. Table 8 lists the indicators related to energy, indoor environment, and comfort and their reasonable ranges. The quantity and quality of daylight transmitted through the windows into the interior space control the visual and color comfort of the occupants.¹⁶⁶ The evaluation indexes of the daylight environment and visual comfort are UDI,¹⁶⁷ daylight glare probability (DGP),¹⁰⁸ Daylight Glare Index (DGI),¹⁶⁸ illuminance uniformity (U_o),¹⁴² correlated color temperature (CCT),¹⁶⁹ and color rendering index (CRI).¹⁷⁰ The solar radiation through the windows affects the indoor thermal environment and occupant thermal comfort, involving indicators such as indoor air temperature (T_a), PMV and PPD,⁷² and the Discomfort index

Table 8. Comprehensive evaluation of indicators and their correlations for smart windows

Indicator	Energy consumption	Thermal environment and thermal comfort	Light environment and visual comfort	Requirement
K	✓	✓		adjustable on the smallest possible basis
T_{lum}	✓	✓	✓	as large as possible
SHGC	✓	✓	✓	larger in cold conditions, smaller in hot conditions
LSG	✓	✓	✓	smaller in cold conditions, larger in hot conditions
UDI			✓	100–2,000 lux
DGP			✓	≤ 0.35
DGI			✓	≤ 22
U_o			✓	≥ 0.7
CCT			✓	3,000–7,500 K
CRI			✓	> 90
T_a		✓		18°C–26°C
PMV		✓		within ± 0.5
PPD		✓		$\leq 10\%$
DI		✓		$15.0 \leq DI \leq 19.9$

(DI).¹⁷¹ The application of TCWs and ECWs should strive to meet the conditions of these indicators to create an energy-saving, comfortable, and healthy environment.

OUTLOOK

This paper reviews the advanced technologies and development of TCWs and ECWs in materials and further illustrates the performance of these two types of windows in buildings by reviewing existing experimental and simulation studies. In addition, we discuss the performance of the ideal smart window and describe the performance distribution of existing smart window technologies by K and LSG. Remarkable achievements and progress have been made in the research of smart windows in recent years, but there are still some obstacles to be solved for wide applications in the future.

The ideal smart window should be able to achieve independent regulation of K and LSG based on large T_{lum} and small K. Future technology should continually strive to increase the T_{lum} of the smart window and enhance the adjustability of the LSG with advanced technology for dual-band modulation. In addition, the core logic of TCWs and ECWs is different. The suitability of TCWs depends heavily on the climate, and the performance of ECWs is closely related to the control strategies. However, as new technologies for dynamic façades emerge, both should be dedicated to improving dynamic regulation performance and enhancing the focus on comprehensive performance.

The eventual application of smart windows places demands on materials and buildings. For materials, it is not only necessary to explore materials with excellent performance but also to ensure the stability, life, and cost reduction of materials. For buildings, it is necessary to evaluate the indoor environment and human comfort after installing smart windows. Application evaluation can be used to guide window performance improvements in reverse.

In summary, the roles of materials and buildings are intersecting and mutually promoting in the development path of smart windows. From material synthesis and technological advances to building-related experiments and simulations, all are aimed at the final application. This paper aims to promote close communication

and cooperation between the two disciplines, hoping to provide suggestions and solid references for future development of smart windows.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.xcrp.2023.101370>.

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AUTHOR CONTRIBUTIONS

S.W. conceived the content and wrote the original manuscript. H.S. guided the framework and writing method. M.D. and H.M. wrote the manuscript and revised the grammar. Y.W. and H.Z. collected data and participated in charting. B.L. provided ideas for the paper, guided the framework, and provided financial support. All authors read and approved the final manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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